A Polynomial Distance-Multiplicity Gap for Every Large Cardinality

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Abstract

For a finite set $A \subset \mathbb{R}^2$, let $f_A(r)$ denote the number of unordered pairs at distance r, and let $G(n)$ be the maximum gap between the two largest values of f_A, taken over all n-point sets. We prove that there is an effectively computable constant $c > 0$ such that $G(n) \geq n^{1+c}$ for every sufficiently large integer n. This proof was realized using assistance of GPT 5.6 Sol.

1 Statement and arithmetic input

For a finite planar set A, let

$$a_1(A) \geq a_2(A) \geq \cdots$$

be the values $f_A(r)$ as r ranges over the distinct positive distances determined by A, listed in nonincreasing order, with zeros appended when necessary. Define

$$\operatorname{gap}(A) = a_1(A) - a_2(A), \qquad G(n) = \max_{|A|=n} \operatorname{gap}(A).$$

Estimating $G(n)$ is Erdős Problem #959 [1]. The present paper does not determine the correct order of growth of $G(n)$, and hence does not resolve that problem in full; it does, however, establish a polynomially superlinear lower bound.

Theorem 1.1. *There is an effectively computable absolute constant $c > 0$ such that $G(n) \geq n^{1+c}$ for every sufficiently large integer n.*

The proof proceeds in four stages. First, Proposition 1.2 extracts fields of every sufficiently large power-of-two degree from Sawin's CM towers [2] while retaining the required local and discriminant bounds. Next, the arithmetic construction isolates a single relative-norm shell from all competitors. Gaussian sampling and a simultaneous concentration argument then produce finite high-dimensional blocks with an exponential multiplicity gap. Finally, planar projection, generic placement, replication, and padding convert these blocks into sets of every sufficiently large cardinality.

Set

$$\mathcal{T} = \{3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43\},$$

$$\mathcal{S} = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 47, 71, 79, 97, 101, 107, 109, 139, 151, 163, 167, 179\},$$

and

$$\lambda = \sqrt{4 \prod_{q \in \mathcal{T}} q}.$$

Proposition 1.2 (Sawin's CM tower, specialized)**.** *Every sufficiently large power of two occurs as the degree $d = [F : \mathbb{Q}]$ of a Galois totally real field F such that, for $K = F(i)$:*

(i) every prime of F above a rational prime in $\mathcal{S}$ splits in K/F;

(ii) for $p \in \mathcal{S}$, the ramification index in $F/\mathbb{Q}$ is 2 when $p \in \{2\} \cup \mathcal{T}$ and 1 otherwise, while the inertia degree is at most 2;

(iii) K/F is unramified at finite places and

$$|\Delta_K|^{1/(2d)} = \lambda;$$

(iv) with $h^-(K) = h(K)/h(F)$,

$$h^-(K) \le 8\lambda^2 \left(\sqrt{\lambda} \log \lambda \frac{e}{4\pi}\right)^d.$$

The result follows by specializing Sawin's Lemma 12 and its proof to the displayed sets $\mathcal{T}$ and $\mathcal{S}$, combining this specialization with the relative class-number estimate in Lemma 9, and applying the power-of-two refinement in Remark 13 [2, Lemmas 9 and 12; Remark 13]. We record the short group-theoretic extraction and the discriminant bookkeeping because the later argument uses this precise formulation.

Put

$$Q_{\mathcal{T}} = \mathbb{Q}\left(\sqrt{\prod_{q \in \mathcal{T}} q}\right), \qquad Q_0 = \mathbb{Q}(\sqrt{q} : q \in \mathcal{T}).$$

Thus $Q_{\mathcal{T}} \subset Q_0$. In Sawin's construction, the fields F are unramified over $Q_{\mathcal{T}}$ and contain Q_0.

Lemma 1.3 (Power-of-two extraction)**.** *Let $F/\mathbb{Q}$ be one of the Galois totally real fields obtained from Sawin's construction, and write*

$$G = \mathrm{Gal}(F/\mathbb{Q}), \qquad N = \mathrm{Gal}(F/Q_0).$$

If $N \neq 1$, then there is an intermediate field E with

$$Q_0 \subset E \subset F, \qquad [E : \mathbb{Q}] = \frac{1}{2}[F : \mathbb{Q}],$$

such that $E/\mathbb{Q}$ is Galois. Iterating this construction produces every sufficiently large power-of-two degree.

Proof. The group G is a finite 2-group and $N \triangleleft G$ is nontrivial. Let G act on N by conjugation. Every nontrivial orbit has even cardinality, so

$$|N| \equiv |N \cap Z(G)| \pmod 2.$$

Since $|N|$ is even and the identity lies in $N \cap Z(G)$, the latter set has at least two elements. Choose a nonidentity element of $N \cap Z(G)$. If necessary, replace it by the element of order two in the cyclic subgroup it generates, and call the resulting central involution z. Then $\langle z \rangle \triangleleft G$, so its fixed field

$$E = F^{\langle z \rangle}$$

is Galois over $\mathbb{Q}$. Because $z \in N$, the field E contains Q_0, and $[E : \mathbb{Q}] = [F : \mathbb{Q}]/2$.

After replacing F by E, the new kernel over Q_0 is $N/\langle z \rangle$. Repeating the argument therefore halves the degree until Q_0 is reached. Since the original finite quotients have arbitrarily large 2-power order, every sufficiently large power of two is obtained. $\square$

Proof of Proposition 1.2. Sawin verifies that the displayed sets $\mathcal{T}$ and $\mathcal{S}$ satisfy the hypotheses of his Lemma 12 [2, Lemma 12]. That lemma and its proof yield Galois totally real fields $F/\mathbb{Q}$ of arbitrarily large degree that contain Q_0 and are unramified over $Q_{\mathcal{T}}$, with the splitting, ramification, and inertia properties in (i)–(ii). The same proof shows that $F(i)/F$ is unramified at every finite place and that

$$\left(\frac{|\Delta_{F(i)}|}{|\Delta_F|}\right)^{1/[F:\mathbb{Q}]} = \sqrt{4\prod_{q\in\mathcal{T}} q} = \lambda.$$

Apply Lemma 1.3. Every intermediate field E produced there contains Q_0 and remains unramified over $Q_{\mathcal{T}}$. Hence the local argument in the proof of Sawin's Lemma 12 applies unchanged: the ramification indices in (ii) are preserved, inertia degrees can only decrease, every prime above $\mathcal{S}$ splits after adjoining i, and $E(i)/E$ is unramified at finite places.

Let $d_E = [E:\mathbb{Q}]$. Since $E/Q_{\mathcal{T}}$ and $E(i)/E$ are unramified at finite places,

$$|\Delta_E| = |\Delta_{Q_{\mathcal{T}}}|^{d_E/2} = \left(4\prod_{q\in\mathcal{T}} q\right)^{d_E/2}, \qquad |\Delta_{E(i)}| = |\Delta_E|^2.$$

Consequently

$$|\Delta_{E(i)}|^{1/(2d_E)} = \sqrt{4\prod_{q\in\mathcal{T}} q} = \lambda,$$

which proves (iii). Finally, Sawin's Lemma 9 [2, Lemma 9], based on Louboutin's relative class-number bound [3, Corollary 3], gives for every such E and $K = E(i)$

$$h^-(K) \le 8\lambda^2\left(\sqrt{\lambda}\log\lambda\frac{e}{4\pi}\right)^{d_E}.$$

This proves (iv). Renaming E as F completes the proof. □

We refer to the degrees supplied by Proposition 1.2 as available tower degrees.

2 Arithmetic shell separation

We now construct a distinguished relative-norm shell and show that a suitable multiplicative weight separates it uniformly from every competing shell.

Let σ denote complex conjugation in K/F, and write

$$F_{>0} = \{\gamma \in F^\times : \tau(\gamma) > 0 \text{ for every real embedding } \tau : F \hookrightarrow \mathbb{R}\}.$$

Define the relative class group $\mathscr{G}_K$ as the set of pairs (J,u), where J is a fractional ideal of K and $u \in F^\times$ is a generator of $\mathrm{N}_{K/F}(J)$, modulo

$$(J,u) \sim ((a)J, a\sigma(a)u), \qquad a \in K^\times.$$

The relevant norm-class sequence is

$$\mathcal{O}_K^\times \xrightarrow{\mathrm{N}_{K/F}} \mathcal{O}_F^\times \longrightarrow \mathscr{G}_K \longrightarrow \mathrm{Cl}(K) \xrightarrow{\mathrm{N}_{K/F}} \mathrm{Cl}(F).$$

The unit quotient has order at most 2^{d+1}, and consequently

$$|\mathscr{G}_K| \le 2^{d+1}h^-(K) \le 16\lambda^2 e^{\kappa d}, \qquad \kappa = \log\left(2\sqrt{\lambda}\log\lambda\frac{e}{4\pi}\right). \tag{1}$$

Fix

$$\eta = \frac{659}{50000}, \qquad s = \frac{2\eta}{\pi}, \qquad k_\eta(Q) = \left\lfloor \frac{1}{Q^\eta - 1} \right\rfloor. \tag{2}$$

For each field in the tower, let

$$\mathfrak{A} = \prod_{\substack{\mathfrak{p} \subset \mathcal{O}_F \\ \mathfrak{p} \text{ split in } K/F}} \mathfrak{p}^{k_\eta(\mathrm{N}\mathfrak{p})}. \tag{3}$$

Only prime ideals with norm at most $2^{1/\eta}$ occur. For each split prime ideal occurring in $\mathfrak{A}$, choose one of the two primes of K above it with the prescribed exponent, and let $J_0 \subset \mathcal{O}_K$ be their product. Then $\mathrm{N}_{K/F}(J_0) = \mathfrak{A}$.

For an integral ideal $\mathfrak{B} \subset \mathcal{O}_F$, let

$$a_K(\mathfrak{B}) = \#\{J \subset \mathcal{O}_K : J \text{ integral and } \mathrm{N}_{K/F}(J) = \mathfrak{B}\}.$$

The ideals of relative norm $\mathfrak{A}$ determine the occupied classes $\mathcal{C}(\mathfrak{A}) \subset \mathscr{G}_K$ via $J \mapsto [JJ_0^{-1}, 1]$. Put $m = |\mathcal{C}(\mathfrak{A})|$. Each occupied class has a representative of the form $[JJ_0^{-1}, 1]$. For every occupied class, choose a representative (J_g, u_g) with u_g totally positive, and set

$$I_g = J_g^{-1} J_0^{-1}, \qquad \alpha_g = u_g^{-1}.$$

Then $\mathrm{N}_{K/F}(I_g) = \mathfrak{A}^{-1}(\alpha_g)$.

For $\gamma \in F_{>0}$ define

$$\mathcal{S}_g(\gamma) = \{\beta \in I_g : \beta\sigma(\beta) = \gamma\alpha_g\}.$$

Let w_K be the number of roots of unity in K.

Proposition 2.1 (Exact target count and uniform competitor bound)**.** *If some $\mathcal{S}_g(\gamma)$ is nonempty, then $\mathfrak{B} = (\gamma)\mathfrak{A}$ is integral and*

$$\sum_g |\mathcal{S}_g(\gamma)| \le w_K a_K(\mathfrak{B}). \tag{4}$$

For the target shell,

$$\sum_g |\mathcal{S}_g(1)| = w_K a_K(\mathfrak{A}). \tag{5}$$

Proof. For $\beta \in \mathcal{S}_g(\gamma)$, the ideal

$$J = (\beta) J_g J_0$$

is integral and has relative norm $(\gamma)\mathfrak{A}$. Its occupied class is recovered from J by

$$g = [JJ_0^{-1}, \gamma],$$

so the contributions from distinct occupied classes are disjoint. Conversely, every ideal J of relative norm $(\gamma)\mathfrak{A}$ whose recovered class is occupied has the form $(\beta)J_gJ_0$ for a solution β. Two solutions give the same ideal exactly when their quotient is a unit of relative norm one. In a CM extension, this kernel is the finite group of roots of unity, of order w_K. This proves (4). When $\gamma = 1$, every ideal of relative norm $\mathfrak{A}$ has an occupied class by definition; hence equality holds in (5). □

Proposition 2.2 (Multiplicative isolation)**.** *There is an effective constant $\rho \in (0, 1)$, depending only on η, such that for every integral $\mathfrak{B} \ne \mathfrak{A}$,*

$$\frac{a_K(\mathfrak{B})}{a_K(\mathfrak{A})} \le \rho \left(\frac{\mathrm{N}\mathfrak{B}}{\mathrm{N}\mathfrak{A}}\right)^\eta. \tag{6}$$

Moreover,

$$a_K(\mathfrak{A}) \ge \mathrm{N}(\mathfrak{A})^\eta. \tag{7}$$

If

$$\nu = \frac{1}{d}\log \mathrm{N}(\mathfrak{A}), \qquad \alpha = \frac{1}{d}\log a_K(\mathfrak{A}),$$

then $\nu \le C_\eta$ for an explicit constant depending only on η.

Proof. The function a_K is multiplicative. At a split prime $\mathfrak{p}$ of norm Q, one has $a_K(\mathfrak{p}^e) = e+1$, so the weighted local sequence is

$$b_Q(e) = (e+1)Q^{-\eta e}.$$

Its consecutive ratios decrease strictly, and its unique maximum occurs at $e = k_\eta(Q)$. This maximum is strict for the rational value $\eta = 659/50000$. Indeed, equality of two consecutive weighted values would give

$$Q^{659}(e+1)^{50000} = (e+2)^{50000}.$$

Write $Q = p^f$. Since $\gcd(e+1, e+2) = 1$, unique factorization forces both $e+1$ and $e+2$ to have no prime divisor other than p. Two consecutive nonnegative powers of one prime can differ by one only when

$$e+1 = 1, \qquad e+2 = 2,$$

so $p = 2$ and $e = 0$. The displayed equality would then read $2^{659f} = 2^{50000}$, which is impossible because $659 \nmid 50000$. Consequently, the ratio of the second-largest value of b_Q to its largest value is strictly less than one. This ratio is at most $1/2$ for all sufficiently large Q; the remaining prime-power norms form a finite set.

Because K/F is unramified at finite places, at an inert prime of norm Q one has

$$a_K(\mathfrak{p}^e) = \begin{cases} 1, & e \text{ even}, \\ 0, & e \text{ odd}. \end{cases}$$

The exponent of $\mathfrak{A}$ at such a prime is zero, so every nontrivial inert local deviation has weighted ratio at most $Q^{-2\eta} \le 2^{-2\eta} < 1$. Let $\rho < 1$ be the maximum of $2^{-2\eta}$ and the finitely many exceptional split-prime second-to-first ratios. If $\mathfrak{B} \neq \mathfrak{A}$, at least one local factor is at most ρ, and every other local factor is at most one. Multiplying over all primes gives (6).

Comparing the maximizing exponent at every split prime with $e = 0$ gives $(k_\eta(Q)+1)Q^{-\eta k_\eta(Q)} \ge 1$. Multiplying over the split primes proves (7). Finally, only rational primes $p \le 2^{1/\eta}$ occur. Because $F/\mathbb{Q}$ is Galois, all primes of F above a fixed rational prime p have common ramification index e and residue degree f, and their number is $d/(ef)$. Their total contribution to ν is therefore

$$\frac{1}{d}\frac{d}{ef}\, k_\eta(p^f) f \log p = \frac{k_\eta(p^f)\log p}{e} \le k_\eta(p^f)\log p.$$

Summing over the finitely many possible rational primes and residue degrees gives

$$\nu \le C_\eta, \qquad C_\eta = \sum_{p \le 2^{1/\eta}} \max_{f \ge 1,\, p^f \le 2^{1/\eta}} k_\eta(p^f)\log p.$$

□

The following finite certificate supplies the only numerical input beyond Proposition 1.2. For $p \in \mathcal{S}$, let e_p be the ramification index specified in that proposition and $f_p \in \{1,2\}$ its inertia degree. Put

$$\nu_c(\mathbf{f}) = \sum_{p\in\mathcal{S}} \frac{k_\eta(p^{f_p})\log p}{e_p}, \qquad \alpha_c(\mathbf{f}) = \sum_{p\in\mathcal{S}} \frac{\log(k_\eta(p^{f_p})+1)}{e_p f_p}.$$

Lemma 2.3 (Rate certificate)**.** *For every allowed inertia profile* $\mathbf{f}$*,*

$$\nu \ge \nu_c(\mathbf{f}) > 573.743, \tag{8}$$

$$-\log 2 - \eta - \kappa + \alpha_c(\mathbf{f}) > 7.541, \tag{9}$$

$$\begin{aligned} &-\log 2 - \eta - \kappa + \alpha_c(\mathbf{f}) \\ &\quad - \eta\left(\kappa + \log\frac{\pi}{\eta} - \log\lambda + \nu_c(\mathbf{f})\right) > 0.0046. \end{aligned} \tag{10}$$

The minima occur at the profile $f_p = 2$ for every p.

Proof. The finite verification, including the exact floor checks, interval bounds, and reduction to the worst-case inertia profile, is given in Section A. The accompanying file `rate_certificate.py` reproduces the certificate. □

3 Gaussian ideal-lattice blocks

We now realize the arithmetic shells as Euclidean length shells in ideal lattices.

Choose one complex embedding $\tau_j : K \hookrightarrow \mathbb{C}$ above each real embedding of F. Define

$$\iota_g(\beta) = \left(\frac{\tau_j(\beta)}{\sqrt{\tau_j(\alpha_g)}}\right)_{j=1}^{d}, \qquad \Lambda_g = \iota_g(I_g) \subset \mathbb{C}^d.$$

If $\beta \in \mathcal{S}_g(\gamma)$, then

$$|\iota_g(\beta)_j|^2 = \tau_j(\gamma), \qquad \|\iota_g(\beta)\|^2 = \sum_{j=1}^{d} \tau_j(\gamma). \tag{11}$$

In particular, every target vector has squared length d.

Lemma 3.1 (Lattice parameters)**.** *All Λ_g have covolume*

$$D = 2^{-d}\frac{\sqrt{|\Delta_K|}}{\mathrm{N}(\mathfrak{A})}. \tag{12}$$

Every nonzero $v \in \Lambda_g$ and $\xi \in \Lambda_g^*$ satisfy

$$\|v\|^2 \geq d\mathrm{N}(\mathfrak{A})^{-1/d}, \tag{13}$$

$$\|\xi\|^2 \geq 4d\left(\frac{\mathrm{N}(\mathfrak{A})}{|\Delta_K|}\right)^{1/d}. \tag{14}$$

Proof. Identify $\mathbb{C}^d$ with $\mathbb{R}^{2d}$ using the Euclidean pairing $\langle z, w\rangle = \mathrm{Re}\sum_j z_j\overline{w_j}$. The ordinary ideal lattice of I_g has covolume $2^{-d}\sqrt{|\Delta_K|}\mathrm{N}_K(I_g)$. The coordinate normalization scales the covolume by $\mathrm{N}_F(\alpha_g)^{-1}$, and $\mathrm{N}_K(I_g) = \mathrm{N}(\mathfrak{A})^{-1}\mathrm{N}_F(\alpha_g)$, proving (12).

For $0 \neq \beta \in I_g$, the ideal $(\beta)I_g^{-1}$ is integral, so

$$|\mathrm{N}_{K/\mathbb{Q}}(\beta)| \geq \mathrm{N}_K(I_g) = \mathrm{N}(\mathfrak{A})^{-1}\mathrm{N}_F(\alpha_g).$$

Thus the product of the d squared coordinate norms of $\iota_g(\beta)$ is at least $\mathrm{N}(\mathfrak{A})^{-1}$, and AM–GM gives (13).

For the chosen Euclidean pairing one has

$$\langle z, w\rangle = \frac{1}{2}\mathrm{Tr}_{K/\mathbb{Q}}\big(z\sigma(w)\big).$$

The factor $1/2$ in this identity implies that the dual of the unnormalized ideal lattice is $2\mathfrak{D}_K^{-1}\sigma(I_g)^{-1}$. Since the normalizing diagonal map is self-adjoint, every vector of Λ_g^* has the form

$$\left(2\tau_j(z)\sqrt{\tau_j(\alpha_g)}\right)_{j=1}^{d}, \qquad z \in \mathfrak{D}_K^{-1}\sigma(I_g)^{-1}.$$

For $z \neq 0$, the ideal $(z)\mathfrak{D}_K\sigma(I_g)$ is integral, whence

$$|\mathrm{N}_{K/\mathbb{Q}}(z)|\,|\Delta_K|\,\mathrm{N}_K(I_g) \geq 1.$$

It follows that the product of the squared coordinate norms of the corresponding dual vector is at least $4^d\mathrm{N}(\mathfrak{A})/|\Delta_K|$. A final application of AM–GM proves (14). □

We use the following elementary lattice-Gaussian estimate. If a lattice $\Lambda \subset \mathbb{R}^n$ has minimum length at least μ, then

$$\sum_{\xi \in \Lambda \setminus \{0\}} e^{-\pi \|\xi\|^2/a} \leq \sum_{j \geq 1} (2j+3)^n e^{-\pi j^2 \mu^2/a}. \tag{15}$$

Indeed, disjoint balls of radius $\mu/2$ around lattice points in the ball of radius $(j+1)\mu$ give the counting factor $(2j+3)^n$.

Poisson summation, Equations (8) and (14), and $|\Delta_K|^{1/d} = \lambda^2$ now imply, uniformly in g, $u \in \mathbb{C}^d$, and $a \in \{s/2, s, 3s/2, 2s\}$,

$$\sum_{x \in \Lambda_g} e^{-\pi a \|x+u\|^2} = \frac{a^{-d}}{D}(1 + O(e^{-100d})). \tag{16}$$

In fact, every nonzero dual vector has squared length greater than $4de^{535}$, and (15), with $n = 2d$ and $a \leq 2s$, is $O(e^{-100d})$. The shift contributes only unit-modulus Fourier factors, so the estimate is uniform in u. Completing the square gives

$$\sum_{x \in \Lambda_g} e^{-\pi s(\|x\|^2 + \|x+v\|^2)} = \frac{(2s)^{-d}}{D} e^{-\pi s \|v\|^2/2}(1 + O(e^{-100d})). \tag{17}$$

Retain each $x \in \Lambda_g$ independently with probability $p(x) = e^{-\pi s \|x\|^2}$, and write $Z_{g,x}$ for its indicator. Because the sum of the retention probabilities is finite, the resulting sample is finite almost surely. Define

$$X = \sum_g \sum_{x \in \Lambda_g} Z_{g,x}, \qquad \mathcal{V}_g(\gamma) = \iota_g(\mathcal{S}_g(\gamma)),$$

and define the ordered shell-pair count by

$$Y_\gamma = \sum_g \sum_{v \in \mathcal{V}_g(\gamma)} \sum_{x \in \Lambda_g} Z_{g,x} Z_{g,x+v}. \tag{18}$$

Every summand is well defined because $v \in \Lambda_g$. Equations Equations (16) and (17), together with Proposition 2.1, give

$$\mathbb{E}X = m \frac{s^{-d}}{D}(1 + O(e^{-100d})), \tag{19}$$

$$\mathbb{E}Y_1 = \frac{(2s)^{-d}}{D} e^{-\eta d} w_K a_K(\mathfrak{A})(1 + O(e^{-100d})). \tag{20}$$

Proposition 3.2 (Expected separation)**.** *There is a constant $\delta > 0$, independent of d, such that for all sufficiently large tower degrees and every $\gamma \neq 1$,*

$$\mathbb{E}(Y_1 - Y_\gamma) \geq \delta \mathbb{E}Y_1.$$

Proof. If the competitor shell is empty, the claim is immediate. Otherwise, set $\mathfrak{B} = (\gamma)\mathfrak{A}$. If $\mathfrak{B} \neq \mathfrak{A}$, then Propositions 2.1 and 2.2 and Equation (17) give

$$\frac{\mathbb{E}Y_\gamma}{\mathbb{E}Y_1} \leq \rho \exp\left[-\eta \sum_{j=1}^{d} (\gamma_j - 1 - \log \gamma_j)\right] + O(e^{-100d}) \leq \rho + O(e^{-100d}),$$

where $\gamma_j = \tau_j(\gamma)$ and $x - 1 - \log x \geq 0$. If $\mathfrak{B} = \mathfrak{A}$ and $\gamma \neq 1$, then γ is a nontrivial totally positive unit. Since $|\mathrm{N}_{F/\mathbb{Q}}(\gamma - 1)| \geq 1$, at least one conjugate lies outside $(0, 2)$. Total positivity therefore implies that some conjugate, say $\gamma_1 = x$, satisfies $x \geq 2$. Since $\prod_j \gamma_j = 1$, AM–GM on the remaining conjugates gives

$$\sum_{j=1}^{d} \gamma_j \geq x + (d-1)x^{-1/(d-1)} \geq 2 + (d-1)2^{-1/(d-1)} \geq d + 1 - \log 2.$$

The last inequality follows from $e^{-y} \geq 1 - y$. The ideal count is then at most the target count, so

$$\frac{\mathbb{E}Y_\gamma}{\mathbb{E}Y_1} \leq e^{-\eta(1-\log 2)} + O(e^{-100d}).$$

Take δ smaller than half of $1 - \max\{\rho, e^{-\eta(1-\log 2)}\}$. □

4 A finite sample separating every shell

The preceding proposition separates the shells in expectation. We now truncate the Gaussian sample and show that one finite realization separates all candidate shells simultaneously.

Fix $H = 10$ and restrict to

$$\mathcal{Q}_g = \{x \in \Lambda_g : \|x\| \leq H\sqrt{d}\}.$$

To bound the discarded point mass, split the Gaussian weight into two equal factors and use one as a tail factor. Applying Equation (16) gives, uniformly in g,

$$\frac{\sum_{\|x\|>H\sqrt{d}} e^{-\pi s\|x\|^2}}{\sum_x e^{-\pi s\|x\|^2}} \leq 2^d e^{-\eta H^2 d}(1 + O(e^{-100d})) = O(e^{-0.6d}),$$

because $\eta H^2 - \log 2 = 0.6248528\ldots$. For a target vector v, so that $\|v\|^2 = d$, the contribution from $\|x\| > H\sqrt{d}$ is bounded by

$$e^{-\eta H^2 d} \sum_x e^{-\pi s(\|x\|^2/2 + \|x+v\|^2)}.$$

Completing the square in the remaining sum and comparing with Equation (17) gives the relative bound

$$\left(\frac{4}{3}\right)^d e^{-\eta(H^2-1/3)d}(1 + O(e^{-100d})) = O(e^{-d}),$$

since $\eta H^2 - \log(4/3) - \eta/3 = 1.0259245\ldots$. The same argument applies when the other endpoint lies outside, so a factor of two bounds the total loss. Truncation decreases competitor counts and removes only $O(e^{-d})$ of the target expectation. Thus Proposition 3.2 remains valid, with a smaller δ, for the truncated variables. From now on X and Y_γ denote the finite sums obtained from (18) by restricting both endpoints to the sets $\mathcal{Q}_g$.

Let $M = \sum_g |\mathcal{Q}_g|$. By Equation (13) and a packing argument,

$$M \leq m(1 + 2He^{\nu/2})^{2d} = e^{O(d)}. \tag{21}$$

The candidate set determines at most M^2 displacement shells. A displacement between candidates has length at most $2H\sqrt{d}$, so Equation (11) and AM–GM give

$$\mathrm{N}_F(\gamma)^{1/d} \leq \frac{1}{d}\sum_j \gamma_j \leq 4H^2, \qquad \mathrm{N}_F(\gamma) \leq (4H^2)^d.$$

For a fixed shell vector, changing one vertex indicator affects at most two ordered pairs, according to whether the vertex is the first or second endpoint. Thus it changes $Y_1 - Y_\gamma$ by at most twice the sum of the target and competitor shell sizes. By Propositions 2.1 and 2.2 this is at most

$$4w_K a_K(\mathfrak{A})(4H^2)^{\eta d}. \tag{22}$$

The unit case $\mathfrak{B} = \mathfrak{A}$ satisfies the same bound.

Proposition 4.1 (Simultaneous finite extraction)**.** *For every sufficiently large tower degree, there is a realization of the truncated sample satisfying*

$$\begin{aligned} \tfrac{1}{2}\mathbb{E}X \le X &\le 2\mathbb{E}X, \\ Y_1 &\ge \tfrac{1}{2}\mathbb{E}Y_1, \\ Y_1 - Y_\gamma &\ge \tfrac{\delta}{2}\mathbb{E}Y_1 \qquad (\gamma \ne 1) \end{aligned}$$

simultaneously for every shell determined by the candidates.

Proof. For a fixed competitor, McDiarmid's bounded-differences inequality [4], together with Equation (22), bounds the failure probability by

$$\exp\left(-\frac{(\mathbb{E}(Y_1 - Y_\gamma))^2}{32Mw_K^2 a_K(\mathfrak{A})^2(4H^2)^{2\eta d}}\right).$$

After inserting Equations (19) to (21), the normalized logarithm of the quantity in the outer exponential is at least

$$\begin{aligned} Q_H = 2(\log 2 + \nu - \log\lambda) &- 2\log(2s) - 2\eta - \kappa \\ &- 2\log(1 + 2He^{\nu/2}) - 2\eta\log(4H^2) + o(1). \end{aligned}$$

The function $2\nu - 2\log(1 + 2He^{\nu/2})$ is increasing. Using Equation (8) with $H = 10$ gives, before the $o(1)$ term,

$$Q_H > 527.7.$$

In particular $Q_H > 500$ for all sufficiently large d. Thus each shell has failure probability at most $\exp(-e^{500d})$, and the number of shells is only $e^{O(d)}$. The same estimate gives the required lower bound for Y_1. Finally, X is a sum of independent Bernoulli variables with $\mathbb{E}X \ge e^{500d}$, so Chernoff's inequality gives its two-sided bound with failure probability $\exp(-e^{\Omega(d)})$. A union bound therefore leaves positive probability. □

5 Planar blocks

The first coordinate of ι_g embeds each sampled component injectively into $\mathbb{C} \simeq \mathbb{R}^2$. Generic placement will prevent distances between components from creating additional high-multiplicity shells. We use the following elementary fact.

Lemma 5.1 (Generic placement)**.** *Given finitely many finite planar sets, there are orientation-preserving rigid motions whose images are disjoint, such that every distance joining two distinct components occurs with multiplicity one and no such distance equals an internal distance.*

Proof. Place the components successively. Suppose a finite union P has already been placed and the next component is C. For distinct prospective cross pairs (c_i, p_j) and (c_k, p_ℓ), equality of their squared distances after a rotation R and translation t is

$$|Rc_i + t - p_j|^2 = |Rc_k + t - p_\ell|^2.$$

The coefficient of t in the difference of the two sides vanishes only when $R(c_i - c_k) = p_j - p_\ell$. There are only finitely many rotations with this property, so choose a different orientation. For that orientation, every such cross-distance equality excludes a line in the translation plane. Equality of a new cross distance with a previously occurring distance or with an internal distance of C excludes a circle, and intersection with a previously placed point excludes a point. There are only finitely many constraints, and a finite union of lines, circles, and points cannot cover $\mathbb{R}^2$. A translation outside this union completes the induction. □

Apply Lemma 5.1 to the nonempty sampled components. Because the selected real embedding $\tau_1 : F \hookrightarrow \mathbb{R}$ is injective, distinct shell parameters give distinct squared distances under the first embedding. By Proposition 4.1, the target internal shell exceeds every competing internal shell by a fixed fraction of $\mathbb{E}Y_1$, and every cross-component distance has multiplicity one.

Proposition 5.2 (Uniform blocks)**.** *There are effective constants $p_+ > 0$ and $\ell_- > 0$ such that every sufficiently large available tower degree d produces a finite planar set A_d with*

$$|A_d| \le e^{p_+ d}, \qquad \operatorname{gap}(A_d) \ge |A_d| e^{\ell_- d}. \tag{23}$$

One may take

$$p_+ = \kappa + \log \frac{\pi}{\eta} - \log \lambda + C_\eta + 1, \qquad \ell_- = 3.$$

Proof. Equation (18) counts every nonzero displacement in both orientations, so passing from ordered to unordered pairs divides all internal shell counts by exactly two. By Proposition 4.1, the target internal multiplicity exceeds every competing internal multiplicity by a fixed positive multiple of $\mathbb{E}Y_1$. By Equation (9), this gap grows exponentially, while every cross-component distance has multiplicity one. Consequently, Equations (1), (19) and (20) give

$$\operatorname{gap}(A_d) \gg |A_d| \, 2^{-d} e^{-\eta d} \frac{w_K a_K(\mathfrak{A})}{m}. \tag{24}$$

Define

$$P_0 = \kappa + \log \frac{\pi}{\eta} - \log \lambda + \nu, \qquad L_0 = -\log 2 - \eta - \kappa + \alpha.$$

Then

$$\frac{1}{d} \log |A_d| \le P_0 + o(1), \qquad \frac{1}{d} \log \frac{\operatorname{gap}(A_d)}{|A_d|} \ge L_0 + o(1).$$

Since $\nu \le C_\eta$, the stated p_+ works. Write $\nu = \nu_c(\mathbf{f}) + \nu_u$, where $\nu_u \ge 0$ is the contribution from the uncontrolled split primes. At each such prime, maximality of $(e+1)Q^{-\eta e}$ against $e = 0$ gives $\log(k_\eta(Q) + 1) \ge \eta k_\eta(Q) \log Q$. Summing yields

$$\alpha \ge \alpha_c(\mathbf{f}) + \eta \nu_u.$$

The uncontrolled contribution therefore cancels in $L_0 - \eta P_0$, and Equation (10) gives

$$L_0 - \eta P_0 > 0.0046.$$

Also, Equation (8) gives $P_0 > 571$, and hence $L_0 > 7$. After absorbing the degree-independent factors and the $o(1)$ terms in (24), we may retain the weaker uniform rate $\ell_- = 3$. ☐

6 Proof for every sufficiently large n

The preceding construction supplies blocks only at available tower degrees. Replication and generic padding remove this restriction and reach every sufficiently large cardinality.

Proof of Theorem 1.1. Let p_+ and ℓ_- be as in Proposition 5.2, and set

$$\theta = \frac{1}{4p_+}, \qquad c = \frac{\ell_-}{16p_+} > 0.$$

Fix a sufficiently large integer n. Every interval $[x/2, x]$ contains a power of two. Since every sufficiently large power of two is an available degree, we may therefore choose an available degree d with

$$\frac{\theta}{2} \log n \le d \le \theta \log n. \tag{25}$$

The associated block has size $N = |A_d| \le e^{p_+ d} \le n^{1/4}$.

Let $t = \lfloor n/N \rfloor$. Take t congruent copies of A_d, after normalizing a distance of maximum multiplicity to one, and place them generically as in Lemma 5.1. Every internal multiplicity is multiplied by t, while every cross-copy distance has multiplicity one. If the two largest multiplicities of A_d are $a_1 \ge a_2$, the two largest values after adding the cross-copy distances are ta_1 and $\max\{ta_2, 1\}$. Thus

$$\operatorname{gap}(B) \ge t \operatorname{gap}(A_d) - 1 \ge tNe^{\ell_- d} - 1.$$

Since $tN \ge n - N \ge n/2$, for sufficiently large n this and Equation (25) give

$$\operatorname{gap}(B) \ge \frac{n}{3} e^{\ell_- d} \ge \frac{1}{3} n^{1+\ell_-/(8p_+)}.$$

It remains to reach cardinality exactly n. Add the missing points one at a time. At each step, choose the new point outside the finitely many perpendicular bisectors that would make two new distances equal, the finitely many circles that would make a new distance repeat an old one, and the existing points. Every newly created distance is then distinct and has multiplicity one. Since $t \ge 2$ for large n, the set B already has cross-copy distances of multiplicity one. The added multiplicity-one distances therefore leave its two largest multiplicity values, and hence their gap, unchanged. For sufficiently large n, the factor $1/3$ is absorbed by replacing $\ell_-/(8p_+)$ with $c = \ell_-/(16p_+)$. Hence

$$G(n) \ge n^{1+c}.$$

□

A Verification of the rate certificate

This appendix records the finite computation used in Lemma 2.3. Its role is purely verificational: the main argument uses only the three inequalities stated in that lemma.

A.1 Certified calculation

The assertion follows from a finite calculation using the displayed formulas. Every floor value $k = k_\eta(q)$ is certified exactly by

$$q^{659} k^{50000} \le (k+1)^{50000}, \qquad q^{659}(k+1)^{50000} > (k+2)^{50000}.$$

After range reduction, the usual alternating series for $\log(1+x)$ gives rational interval enclosures for every logarithm. Directed interval arithmetic then gives the following certified bounds at the profile with all entries equal to 2:

quantity	certified interval
ν_c	(573.7430444817, 573.7430444818)
α_c	(19.8003353767, 19.8003353768)
$-\log 2 - \eta - \kappa + \alpha_c$	(7.5418334325, 7.5418334327)
left side of (10)	(0.0046211747, 0.0046211748)

For each controlled prime, the $f = 1$ contribution minus the $f = 2$ contribution is positive. More precisely, the minimum of these twenty-two local differences is greater than 18.5 for ν_c, greater than 0.96 for α_c, and greater than 0.70 for the expression in (10). These bounds are obtained by the same exact floor checks and rational logarithm enclosures. Thus the all-2 profile is componentwise minimal. The three displayed bounds then follow with substantial rounding slack.

A.2 Reproducibility note

The ancillary source includes the script `rate_certificate.py`. It checks every floor value by the two exact integer inequalities displayed above, uses directed interval arithmetic for the transcendental quantities, verifies the worst-case profile bounds, and checks the local monotonicity inequalities for each controlled prime. Thus it independently confirms both the numerical intervals and the reduction from all allowed inertia profiles to the worst-case profile. It can be run from the extracted source directory with `python3 anc/rate_certificate.py`; the script exits with a nonzero status if any certified inequality fails. It requires Python 3 and `mpmath` version 1.3.0 or later.

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